

Richie Tran

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EDUCATION

Carnegie Mellon University

B.S. Computer Science and Machine Learning

Pittsburgh, PA

May 2027

Relevant Coursework

Data Structures and Algorithms, Computer Systems, AI Representation and Problem Solving, Machine Learning I, Discrete Mathematics, Functional Programming, Ideas in Theoretical Computer Science

EXPERIENCE

Software Engineer Intern | *Opportunity Knocks*

October 2025 - Present

- Launched mobile app to 5,000+ users by leading development in React Native and Supabase, translating the existing web application to cross-platform iOS and Android.
- Reduced bug reports by 30% by implementing new features and resolving issues across the production web application.

Software Developer | *Scotty Labs, Carnegie Mellon University*

August 2024 – Present

- Built an AI-powered presentation tool in React and Node.js, enabling 100+ users to receive real-time feedback on delivery.
- Improved campus accessibility for 1,000+ students by contributing to CMUMaps, CMUResearch, and CMUEats.

Driverless Control Engineer | *Carnegie Mellon Racing*

August 2024 – Present

- Reduced path-planning latency by optimizing algorithms in C, integrating sensor fusion for autonomous navigation across multiple track configurations.
- Contributed to team's 13th place finish out of 100 teams at Formula FSAE Electric.

Lead Programmer | *FRC, Clarkstown Cyborgs*

September 2020 – August 2024

- Engineered autonomous path-planning and swerve drive systems, contributing to regional finalist placement at FIRST Robotics Competition.
- Accelerated team development by mentoring 40+ members on software, mechanical, and electrical systems.
- Increased team funding by \$3000 (30%) by developing a team website that improved community engagement.

PROJECTS

MyEyes | *Python, Swift, Flask, CoreML, YOLOv8*

- Developed an audio-based iOS app enabling visually impaired users to navigate safely through real-time hazard detection.
- Achieved 90% detection accuracy by deploying a fine-tuned YOLOv8 model via CoreML with a custom urgency-weighted loss function.
- Integrated LiDAR and camera-based obstacle identification with Siri commands for accessibility.

Dynamic Storage Allocator | *C*

- 64-bit dynamic memory allocator with correct alignment and metadata management.
- Achieved 75% space utilization and 10,000 Kops/sec throughput using a segregated free list with optimized search strategies

Virtual Machine | *C, Bytecode*

- Custom virtual machine and translator pipeline for converting a simplified C-like language into C.
- Implemented a bytecode interpreter with parsing, memory management, and instruction dispatch.

SKILLS

Programming Languages: C, C++, Python, Swift, Java, JavaScript, HTML, CSS, TypeScript, Bash

Tools: Node.js, React, Django, Firebase, Supabase, React Native, SwiftUI, Git, Linux, NumPy, Figma